

Supplementary Information: On the Accuracy of Point Estimation Methods for Moment-Based Reliability and Sensitivity Analysis

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S1 Error computed from RM for independent system of random variables.

$$\begin{aligned}
 \text{Error} = & \sum_{j=4}^{\infty} \frac{1}{j!} \sum_{i=1}^n \frac{\partial^j f(\boldsymbol{\mu})}{\partial \xi_i^j} \left\{ \mathbb{E}(\xi_i - \mu_i)^j - \sigma_i^j \sum_{k_i=1}^2 \omega_{i,k_i} (\eta_{k_i} \alpha_{i,k_i})^j \right\} \\
 & + \sum_{j_1=2}^{\infty} \sum_{j_2=2}^{\infty} \frac{1}{j_1! j_2!} \sum_{i_1 < i_2} \frac{\partial^{j_1+j_2} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1} \partial \xi_{i_2}^{j_2}} \left\{ \mathbb{E}(\xi_{i_1} - \mu_{i_1})^{j_1} \mathbb{E}(\xi_{i_2} - \mu_{i_2})^{j_2} \right. \\
 & - \sigma_{i_1}^{j_1} \sigma_{i_2}^{j_2} \sum_{k_{i_1}=1}^2 \omega_{i_1,k_{i_1}} (\eta_{k_{i_1}} \alpha_{i_1,k_{i_1}})^{j_1} \sum_{k_{i_2}=1}^2 \omega_{i_2,k_{i_2}} (\eta_{k_{i_2}} \alpha_{i_2,k_{i_2}})^{j_2} \left. \right\} \\
 & + \sum_{j_1=2}^{\infty} \sum_{j_2=2}^{\infty} \sum_{j_3=2}^{\infty} \frac{1}{j_1! j_2! j_3!} \sum_{i_1 < i_2 < i_3} \frac{\partial^{j_1+j_2+j_3} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1} \partial \xi_{i_2}^{j_2} \partial \xi_{i_3}^{j_3}} \left\{ \mathbb{E}(\xi_{i_1} - \mu_{i_1})^{j_1} \dots \mathbb{E}(\xi_{i_3} - \mu_{i_3})^{j_3} \right. \\
 & - \sigma_{i_1}^{j_1} \sigma_{i_2}^{j_2} \sigma_{i_3}^{j_3} \sum_{k_{i_1}=1}^2 \omega_{i_1,k_{i_1}} (\eta_{k_{i_1}} \alpha_{i_1,k_{i_1}})^{j_1} \sum_{k_{i_2}=1}^2 \omega_{i_2,k_{i_2}} (\eta_{k_{i_2}} \alpha_{i_2,k_{i_2}})^{j_2} \sum_{k_{i_3}=1}^2 \omega_{i_3,k_{i_3}} (\eta_{k_{i_3}} \alpha_{i_3,k_{i_3}})^{j_3} \left. \right\} \\
 & + \sum_{j_1=2}^{\infty} \dots \sum_{j_n=2}^{\infty} \frac{1}{j_1! \dots j_n!} \sum_{i_1 < \dots < i_n} \frac{\partial^{j_1+\dots+j_n} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1} \dots \partial \xi_{i_n}^{j_n}} \left\{ \mathbb{E}(\xi_{i_1} - \mu_{i_1})^{j_1} \dots \mathbb{E}(\xi_{i_n} - \mu_{i_n})^{j_n} \right. \\
 & - \sigma_{i_1}^{j_1} \dots \sigma_{i_n}^{j_n} \sum_{k_{i_1}=1}^2 \omega_{i_1,k_{i_1}} (\eta_{k_{i_1}} \alpha_{i_1,k_{i_1}})^{j_1} \dots \sum_{k_{i_n}=1}^2 \omega_{i_n,k_{i_n}} (\eta_{k_{i_n}} \alpha_{i_n,k_{i_n}})^{j_n} \left. \right\}. \tag{S1.1}
 \end{aligned}$$

S2 Error computed from HM for independent system of random variables.

$$\begin{aligned}
 \text{Error} = & \sum_{j=3}^{\infty} \frac{1}{j!} \sum_{i=1}^n \frac{\partial^j f(\boldsymbol{\mu})}{\partial \xi_i^j} \left\{ \mathbb{E}(\xi_i - \mu_i)^j - \frac{1}{2} \sigma_i^j n^{j/2-1} (1 + (-1))^j \right\} \\
 & + \sum_{j_1=2}^{\infty} \sum_{j_2=2}^{\infty} \frac{1}{j_1! j_2!} \sum_{i_1 < i_2} \frac{\partial^{j_1+j_2} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1} \partial \xi_{i_2}^{j_2}} \mathbb{E}((\xi_{i_1} - \mu_{i_1})^{j_1} (\xi_{i_2} - \mu_{i_2})^{j_2}) + \dots \\
 & + \sum_{j_1=2}^{\infty} \dots \sum_{j_n=2}^{\infty} \frac{1}{j_1! \dots j_n!} \sum_{i_1 < \dots < i_n} \frac{\partial^{j_1+\dots+j_n} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1} \dots \partial \xi_{i_n}^{j_n}} \mathbb{E}(\xi_{i_1} - \mu_{i_1})^{j_1} \dots \mathbb{E}(\xi_{i_n} - \mu_{i_n})^{j_n}. \tag{S2.1}
 \end{aligned}$$

S3 Error computed from LM for independent system of random variables.

$$\begin{aligned}
\text{Error} &= \sum_{j=5}^{\infty} \frac{1}{j!} \sum_{i=1}^n \frac{\partial^j f(\boldsymbol{\mu})}{\partial \xi_i^j} \left\{ \mathbb{E}(\xi_i - \mu_i)^j - \sigma_i^j (\omega_i^+ (\alpha_i^+)^j + \omega_i^- (\alpha_i^-)^j) \right\} \\
&+ \sum_{j_1=2}^{\infty} \sum_{j_2=2}^{\infty} \frac{1}{j_1! j_2!} \sum_{i_1 < i_2} \frac{\partial^{j_1+j_2} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1} \partial \xi_{i_2}^{j_2}} \mathbb{E}(\xi_{i_1} - \mu_{i_1})^{j_1} \mathbb{E}(\xi_{i_2} - \mu_{i_2})^{j_2} \\
&+ \sum_{j_1=2}^{\infty} \cdots \sum_{j_n=2}^{\infty} \frac{1}{j_1! \cdots j_n!} \sum_{i_1 < \cdots < i_n} \frac{\partial^{j_1+\cdots+j_n} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1} \cdots \partial \xi_{i_n}^{j_n}} \mathbb{E}(\xi_{i_1} - \mu_{i_1})^{j_1} \cdots \mathbb{E}(\xi_{i_n} - \mu_{i_n})^{j_n}.
\end{aligned} \tag{S3.1}$$

S4 Error computed from MHM for independent system of random variables.

$$\begin{aligned}
\text{Error} &= \sum_{j=3}^{\infty} \frac{1}{j!} \sum_{i=1}^n \frac{\partial^j f(\boldsymbol{\mu})}{\partial \xi_i^j} \left\{ \mathbb{E}(\xi_i - \mu_i)^j - \frac{1}{2} \sigma_i^j n^{j/2-1} (1 + (-1)^j) \right\} \\
&+ \sum_{j_1=2}^{\infty} \sum_{j_2=2}^{\infty} \frac{1}{j_1! j_2!} \sum_{i_1 < i_2} \frac{\partial^{j_1+j_2} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1} \partial \xi_{i_2}^{j_2}} \mathbb{E}(\xi_{i_1} - \mu_{i_1})^{j_1} \mathbb{E}(\xi_{i_2} - \mu_{i_2})^{j_2} + \cdots \\
&+ \sum_{j_1=2}^{\infty} \cdots \sum_{j_n=2}^{\infty} \frac{1}{j_1! \cdots j_n!} \sum_{i_1 < \cdots < i_n} \frac{\partial^{j_1+\cdots+j_n} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1} \cdots \partial \xi_{i_n}^{j_n}} \mathbb{E}(\xi_{i_1} - \mu_{i_1})^{j_1} \cdots \mathbb{E}(\xi_{i_n} - \mu_{i_n})^{j_n}.
\end{aligned} \tag{S4.1}$$

S5 Response mean approximated from DRM

Substituting $s = 1$ in Eqn (2.6.1) of the main text and applying expectation operator on its both sides, we get the approximated response mean as:

$$\hat{\mu}_Y = \sum_{i=1}^n \mathbb{E}(f(\mu_1, \dots, \mu_{i-1}, \xi_i, \mu_{i+1}, \dots, \mu_n)) - (n-1)f(\boldsymbol{\mu}). \tag{S5.1}$$

Now v number of concentration points $\xi_{i,k} = \mu_i + \alpha_{i,k} \sigma_i$ with the corresponding weights $\omega_{i,k}$, $k = 1, 2, \dots, v$ are selected for each random variable ξ_i , $i = 1, 2, \dots, n$. The term $f(\mu_1, \dots, \mu_{i-1}, \xi_{i,k}, \mu_{i+1}, \dots, \mu_n)$ is expanded in Taylor's series about the input mean $\boldsymbol{\mu} = (\mu_1, \mu_2, \dots, \mu_n)$. This yields

$$\hat{\mu}_Y = \sum_{i=1}^n \sum_{k=1}^v \omega_{i,k} \left(f(\boldsymbol{\mu}) + \sum_{j=1}^{\infty} \frac{1}{j!} \frac{\partial^j f(\boldsymbol{\mu})}{\partial \xi_i^j} (\alpha_{i,k} \sigma_i)^j \right) - (n-1)f(\boldsymbol{\mu}). \tag{S5.2}$$

It may be noted that $\sum_{k=1}^v \omega_{i,k} = 1$, then

$$\hat{\mu}_Y = f(\boldsymbol{\mu}) + \sum_{j=1}^{\infty} \frac{1}{j!} \sum_{i=1}^n \frac{\partial^j f(\boldsymbol{\mu})}{\partial \xi_i^j} \sigma_i^j \sum_{k=1}^v \omega_{i,k} \alpha_{i,k}^j. \quad (\text{S5.3})$$

Now, let us consider $s = 2$. The estimated mean ($\hat{\mu}_Y$) of the response function is given by:

$$\begin{aligned} \hat{\mu}_Y &= \sum_{i_1 < i_2} \mathbb{E}(f(\mu_1, \dots, \mu_{i_1-1}, \xi_{i_1}, \mu_{i_1+1}, \dots, \mu_{i_2-1}, \xi_{i_2}, \mu_{i_2+1}, \dots, \mu_n)) \\ &- (n-2) \sum_{i=1}^n \mathbb{E}(f(\mu_1, \dots, \mu_{i-1}, \xi_i, \mu_{i+1}, \dots, \mu_n)) + \frac{(n-1)(n-2)}{2} f(\boldsymbol{\mu}). \end{aligned} \quad (\text{S5.4})$$

The expectation terms are obtained using the product rule of numerical integration as follows:

$$\hat{\mu}_Y = \sum_{i_1 < i_2} \sum_{k_1=1}^v \sum_{k_2=1}^v \omega_{i_1,k_1} \omega_{i_2,k_2} f(\xi_{i_1,k_1}, \xi_{i_2,k_2}) - (n-2) \sum_{i=1}^n \sum_{k=1}^v \omega_{i,k} f(\xi_{i,k}) + \frac{(n-1)(n-2)}{2} f(\boldsymbol{\mu}). \quad (\text{S5.5})$$

Here, for our convenience, we ignore the input variables fixed with their mean values. Expanding the terms $f(\xi_{i_1,k_1}, \xi_{i_2,k_2})$ and $f(\xi_{i,k})$ in Taylor's series about the input mean $\boldsymbol{\mu} = (\mu_1, \mu_2, \dots, \mu_n)$, we get

$$\begin{aligned} \hat{\mu}_Y &= f(\boldsymbol{\mu}) + \sum_{j=1}^{\infty} \frac{1}{j!} \sum_{i=1}^n \frac{\partial^j f(\boldsymbol{\mu})}{\partial \xi_i^j} \sigma_i^j \sum_{k=1}^v \omega_{i,k} \alpha_{i,k}^j \\ &+ \sum_{j_1=1}^{\infty} \sum_{j_2=1}^{\infty} \frac{1}{j_1! j_2!} \sum_{i_1 < i_2} \frac{\partial^{j_1+j_2} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1} \partial \xi_{i_2}^{j_2}} \sigma_{i_1}^{j_1} \sigma_{i_2}^{j_2} \sum_{k_1=1}^v \omega_{i_1,k_1} \alpha_{i_1,k_1}^{j_1} \sum_{k_2=1}^v \omega_{i_2,k_2} \alpha_{i_2,k_2}^{j_2}. \end{aligned} \quad (\text{S5.6})$$

S6 Error computed from MRM for independent system of random variables.

$$\begin{aligned} \text{Error} &= \sum_{j=4}^{\infty} \frac{1}{j!} \sum_{i=1}^n \frac{\partial^j f(\boldsymbol{\mu})}{\partial \xi_i^j} \{ \mathbb{E}(\xi_i - \mu_i)^j - \sigma_i^j (\omega_i^+ (\alpha_i^+)^j + (-1)^j \omega_i^- (\alpha_i^-)^j) \} \\ &+ \sum_{j_1=2}^{\infty} \sum_{j_2=2}^{\infty} \frac{1}{j_1! j_2!} \sum_{i_1 < i_2} \frac{\partial^{j_1+j_2} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1} \partial \xi_{i_2}^{j_2}} \mathbb{E}(\xi_{i_1} - \mu_{i_1})^{j_1} \mathbb{E}(\xi_{i_2} - \mu_{i_2})^{j_2} \\ &+ \sum_{j_1=2}^{\infty} \dots \sum_{j_n=2}^{\infty} \frac{1}{j_1! \dots j_n!} \sum_{i_1 < \dots < i_n} \frac{\partial^{j_1+\dots+j_n} f(\boldsymbol{\mu})}{\partial \xi_{i_1}^{j_1} \dots \partial \xi_{i_n}^{j_n}} \mathbb{E}(\xi_{i_1} - \mu_{i_1})^{j_1} \dots \mathbb{E}(\xi_{i_n} - \mu_{i_n})^{j_n}. \end{aligned} \quad (\text{S6.1})$$